RESEARCH PAPER

Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor

Unified Mentor Internship
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Live dashboard: https://factorytocustomershippingrouteefficiency.streamlit.app/

Important note on the data

The shipping lead times in this dataset are not realistic. Orders were placed between January 2024 and December 2025, but ship dates fall between June 2026 and June 2030, giving lead times of roughly 900 to 1,640 days. The ship dates appear to be synthetic or shifted.

All findings in this paper should therefore be read as relative comparisons between routes, states and ship modes. They are not estimates of real-world delivery times.

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Abstract

Nassau Candy Distributor ships products from five factories to customers in the United States and Canada, but its logistics decisions are not supported by route-level efficiency measures. This paper analyses 10,194 order lines to rank factory-to-customer routes, locate geographic bottlenecks and compare ship modes. Routes are defined as Factory to customer Region and Factory to customer State, and are scored on average lead time, lead time variability, delay frequency and a normalized Route Efficiency Score.

Regional differences are small (about 12 days between the fastest and slowest region), while state-level differences are larger (about 217 days among states with at least 30 orders). Tennessee and Indiana stand out as high-volume states that are slower than average on every measure we tested. Expedited ship modes were not faster than Standard Class, and cost per unit and profit margin were nearly identical across modes.

Data limitation. The lead times in the dataset are not realistic (about 904 to 1,642 days). We also found that lead time depends strongly on the year the order was placed (2024 orders average 1,094 days, 2025 orders average 1,479 days), which distorts raw state and delay comparisons. We therefore report a year-adjusted check alongside the raw results and treat all conclusions as relative, not absolute.

Keywords: shipping lead time, route efficiency, logistics analytics, geographic bottlenecks, ship mode, Streamlit.

1. Introduction

Shipping efficiency directly affects customer satisfaction, operating cost and a distributor's ability to scale. Nassau Candy Distributor holds rich order and shipment records, yet it lacks clear answers to four practical questions: which factory-to-customer routes are consistently efficient, which routes are frequently delayed, how performance varies by region, state and ship mode, and where operational bottlenecks sit geographically. Without these answers, logistics improvement stays reactive.

This project turns the raw order data into route-level operational intelligence. It has three outputs: this research paper, an interactive Streamlit dashboard, and an executive summary for stakeholders.

1.1 Objectives

- Define and measure shipping lead time for every order.
- Rank routes from fastest to slowest and identify the 10 most and least efficient routes.
- Identify regions and states with high lead times, or high volume combined with poor performance.
- Compare Standard and Expedited shipping, including a descriptive cost-time view.
- Recommend practical, data-driven next steps.

2. Data and Methodology

2.1 Dataset

The dataset contains 10,194 order lines and 18 fields, including order and ship dates, ship mode, customer geography, product, sales, units, gross profit and cost. There are no missing values and no duplicate rows. Orders span 2 January 2024 to 31 December 2025. Customers are located in 59 states and provinces: 9,994 order lines are in the United States and 200 are in Canada. The business is dominated by chocolate (9,844 order lines, 96.6%); the Other division has 310 and Sugar has 40. Total sales are $141,783.63 with an overall gross profit margin of 65.9%.

The dataset does not include a factory field. We derived the factory by mapping each product to its manufacturing factory using the product-factory table in the project specification, and we used the factory coordinates from the same specification. All 15 products mapped successfully.

Table 1. Orders by factory

Factory	Order lines	Avg lead time (days)	Std dev (days)	Sales ($)
Lot's O' Nuts	5,692	1,321.2	261.6	76,340
Wicked Choccy's	4,152	1,321.1	262.2	55,353
Secret Factory	217	1,321.9	289.1	8,588
The Other Factory	100	1,280.3	259.7	1,282
Sugar Shack	33	1,340.0	265.6	221

Two factories account for 96.6% of all order lines. Routes from the other three factories have very few orders, so their route-level results are statistically weak.

2.2 Data cleaning and validation

- Order and ship dates were parsed from DD-MM-YYYY format. No dates failed to parse.
- Lead time was computed as Ship Date minus Order Date. No lead times were negative, so no rows were removed.
- Text fields (city, state, region, ship mode, division) were trimmed and standardized. No duplicates or missing values were found.
- Canadian provinces were kept in the tables but excluded from the US map.

2.3 Feature engineering and route definition

Each order line was assigned a factory, a Factory to Region route and a Factory to State route. Ship modes were also grouped into Standard (Standard Class) and Expedited (First Class, Second Class, Same Day). Profit margin and cost per unit were computed for the cost-time comparison. This gives 20 region-level routes and 196 state-level routes.

2.4 Key performance indicators

Table 2. KPI definitions

KPI	Definition
Shipping Lead Time	Ship Date minus Order Date, in days
Average Lead Time	Mean lead time per route
Route Volume	Number of order lines per route
Lead Time Variability	Standard deviation of lead time per route
Delay Frequency	Share of shipments with lead time above the threshold (75th percentile, 1,638 days)
Route Efficiency Score	Min-max normalized average lead time: 100 for the fastest route, 0 for the slowest

Routes with fewer than 10 order lines were excluded from rankings because averages from very small samples are unreliable. This leaves 17 of 20 region-level routes and 96 of 196 state-level routes. For state-level geographic analysis we used states with at least 30 order lines (40 states).

3. Data Quality and Limitations of the Lead Time Data

Before interpreting any result, the reader should understand the main weakness of the data.

3.1 Lead times are not realistic

Order dates run from January 2024 to December 2025. Ship dates run from June 2026 to June 2030. As a result, every lead time is between 904 and 1,642 days (mean 1,320.8, median 1,274). Real shipments of consumer goods take days, not years. The ship dates are therefore best treated as synthetic or shifted, and the numbers cannot be read as delivery times.

3.2 Lead time is driven by order year

The lead time distribution is not a smooth curve. It forms three separate clusters (Figure 1): about 2,051 order lines at roughly 900 to 1,100 days, 4,764 at roughly 1,200 to 1,300 days, and 3,379 at roughly 1,600 to 1,700 days. No shipment falls between the clusters.

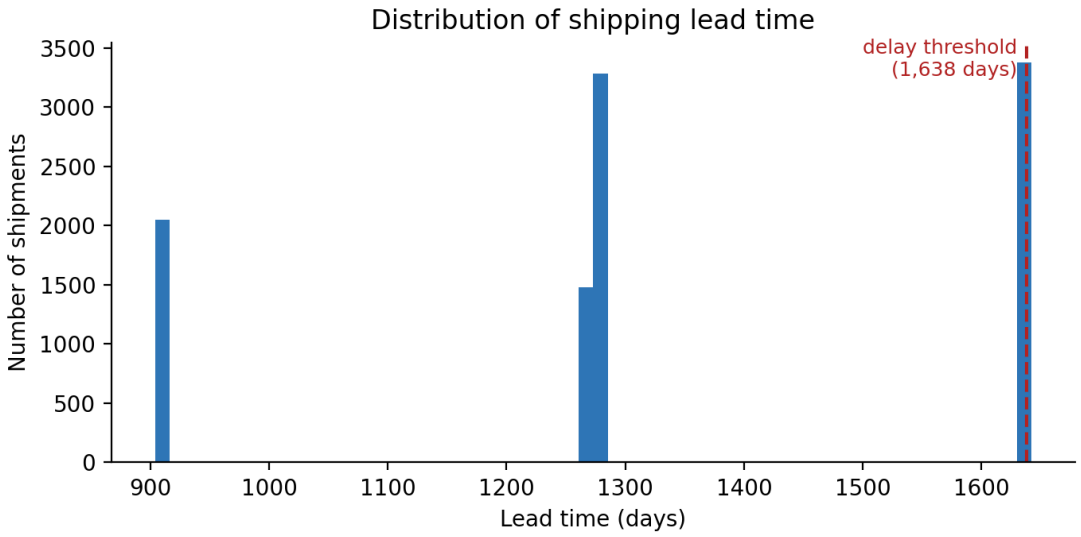


Figure 1. Distribution of lead time (days). Three separate clusters are visible.

These clusters line up with the order year. Orders placed in 2024 have an average lead time of 1,094 days and no shipment above the delay threshold, while orders placed in 2025 average 1,479 days and 36% of them exceed the threshold. The spread inside each year is nearly the same (standard deviation about 182 days in both). Figure 2 shows the step change between 2024 and 2025.

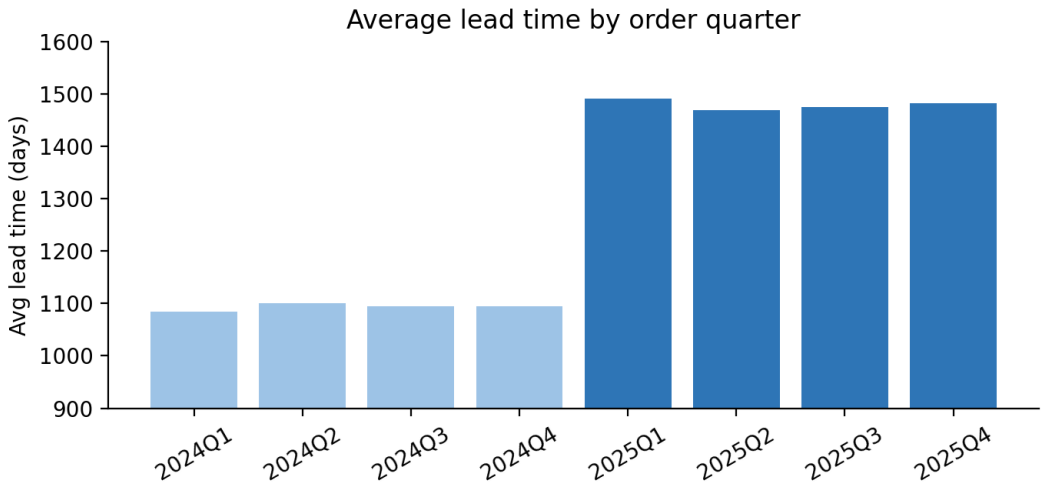


Figure 2. Average lead time by order quarter. Lead time jumps in the first quarter of 2025.

This matters for three reasons. First, any state or route with a larger share of 2025 orders will look slower for reasons unrelated to its geography. Second, the delay frequency KPI is largely a flag for 2025 orders. Third, the ship mode comparison cannot be interpreted as a true speed comparison.

How this paper handles the limitation

1. Raw results are reported as the brief requires, and they are described as relative comparisons only.

2. A year-adjusted check (Section 8) subtracts each order's own-year average lead time, so that differences between states are not driven by order-year mix.

3. Recommendations are phrased as relative priorities and are conditional on the real shipping dates being verified.

4. Exploratory Data Analysis

The typical order is small: the mean order value is $13.91 with 3.79 units, a mean gross profit of $9.17 and a mean cost of $4.74. The largest single order is $260. Lead time has a 5th percentile of 907 days, a median of 1,274 days and a 95th percentile of 1,641 days (standard deviation 262.4).

Average lead time is almost identical across the three large dimensions we looked at. By factory it ranges from 1,280 days (The Other Factory, 100 orders) to 1,340 days (Sugar Shack, 33 orders), but the two factories that carry almost all of the volume are within 0.2 days of each other (1,321.2 and 1,321.1). By region it ranges from 1,311 to 1,323 days. By ship mode it ranges from 1,314 to 1,338 days. The exploratory pattern is therefore that differences exist mostly at the state and route level, not in broad groupings.

Variability is high everywhere. Standard deviations of 250 to 290 days appear for almost every group, meaning that individual orders on the same route differ widely and the average alone is an incomplete summary. Much of this variability reflects the order-year effect described in Section 3.

5. Route Efficiency Results

5.1 Region-level routes

Among the 17 region-level routes with at least 10 orders, the fastest is The Other Factory to Interior (1,206.6 days, but only 11 orders) and the slowest is Sugar Shack to Atlantic (1,375.4 days, 18 orders). Both rest on small samples. Among routes with at least 500 orders, the fastest is Wicked Choccy's to Gulf (1,309.1 days, 663 orders) and the slowest is Wicked Choccy's to Pacific (1,329.1 days, 1,342 orders), a gap of only 20 days.

Table 3. Selected region-level routes (10+ orders), sorted by efficiency score

Route	Orders	Avg lead (days)	Std dev	Delay %	Score
The Other Factory to Interior	11	1,206.6	274.4	0.0	100.0
The Other Factory to Gulf	19	1,272.6	272.7	15.8	60.9
Wicked Choccy's to Gulf	663	1,309.1	259.9	20.4	39.3
Lot's O' Nuts to Gulf	897	1,314.0	266.6	20.3	36.4
Lot's O' Nuts to Pacific	1,813	1,317.5	266.7	20.7	34.3
Lot's O' Nuts to Atlantic	1,661	1,326.8	256.0	20.7	28.8
Wicked Choccy's to Pacific	1,342	1,329.1	266.2	22.3	27.4

Route	Orders	Avg lead (days)	Std dev	Delay %	Score
Secret Factory to Atlantic	72	1,349.2	280.7	25.0	15.5
Sugar Shack to Atlantic	18	1,375.4	274.6	33.3	0.0

5.2 State-level routes

There are 96 state-level routes with at least 10 orders. The 10 fastest and 10 slowest are shown in Figure 3 and Tables 4 and 5. The fastest route is Wicked Choccy's to Nevada (1,182.2 days, 12 orders) and the slowest is Wicked Choccy's to New Mexico (1,488.9 days, 17 orders), a spread of about 307 days. Several of the extreme routes have fewer than 20 orders, so we place more weight on routes with larger volume. Lot's O' Nuts to Virginia (109 orders) and Wicked Choccy's to Virginia (106 orders) are fast routes with solid volume. Lot's O' Nuts to Tennessee (109 orders), Wicked Choccy's to Tennessee (67 orders) and Wicked Choccy's to Indiana (54 orders) are slow routes with solid volume.

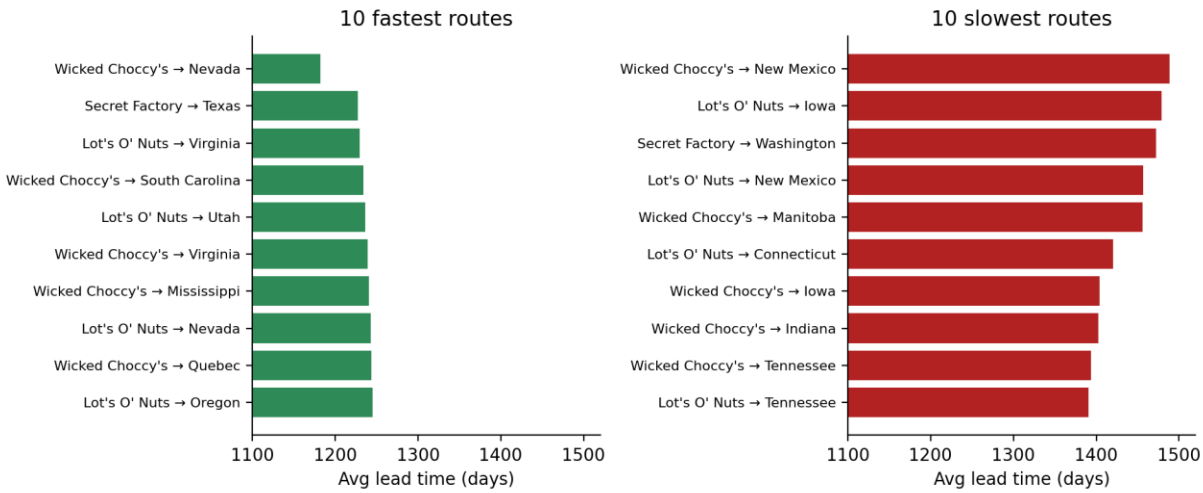


Figure 3. The 10 fastest and 10 slowest state-level routes (10+ orders).

Table 4. Ten most efficient state-level routes

Route	Orders	Avg lead (days)	Std dev	Delay %
Wicked Choccy's to Nevada	12	1,182.2	226.4	8.3
Secret Factory to Texas	16	1,227.1	323.9	18.8
Lot's O' Nuts to Virginia	109	1,229.5	262.0	13.8
Wicked Choccy's to South Carolina	19	1,234.2	240.9	10.5
Lot's O' Nuts to Utah	29	1,236.0	298.2	20.7
Wicked Choccy's to Virginia	106	1,238.9	254.8	12.3
Wicked Choccy's to Mississippi	22	1,240.5	222.9	9.1
Lot's O' Nuts to Nevada	24	1,243.2	239.2	16.7
Wicked Choccy's to Quebec	24	1,243.6	322.1	29.2

Route	Orders	Avg lead (days)	Std dev	Delay %
Lot's O' Nuts to Oregon	77	1,245.2	269.9	16.9

Table 5. Ten least efficient state-level routes

Route	Orders	Avg lead (days)	Std dev	Delay %
Lot's O' Nuts to Tennessee	109	1,391.0	257.7	31.2
Wicked Choccy's to Tennessee	67	1,393.8	241.4	32.8
Wicked Choccy's to Indiana	54	1,402.5	257.8	42.6
Wicked Choccy's to Iowa	14	1,403.8	272.0	28.6
Lot's O' Nuts to Connecticut	47	1,420.6	249.4	36.2
Wicked Choccy's to Manitoba	10	1,455.6	192.9	40.0
Lot's O' Nuts to New Mexico	18	1,456.9	258.3	55.6
Secret Factory to Washington	11	1,472.4	190.5	18.2
Lot's O' Nuts to Iowa	16	1,479.0	187.5	37.5
Wicked Choccy's to New Mexico	17	1,488.9	226.4	64.7

The same destination often appears for both big factories (Virginia fast for both, Tennessee and Iowa and New Mexico slow for both). This points to a destination-side pattern rather than a factory-side pattern, which fits the finding that the two large factories have nearly identical average lead times.

6. Geographic Bottleneck Analysis

6.1 Regions

The four customer regions perform almost identically (Figure 4 and Table 6). Gulf is the fastest at 1,311.4 days and Interior the slowest at 1,323.1 days, a gap of about 12 days. Delay frequency ranges from 20.5% to 21.5%. Regions are therefore not where the bottlenecks are.

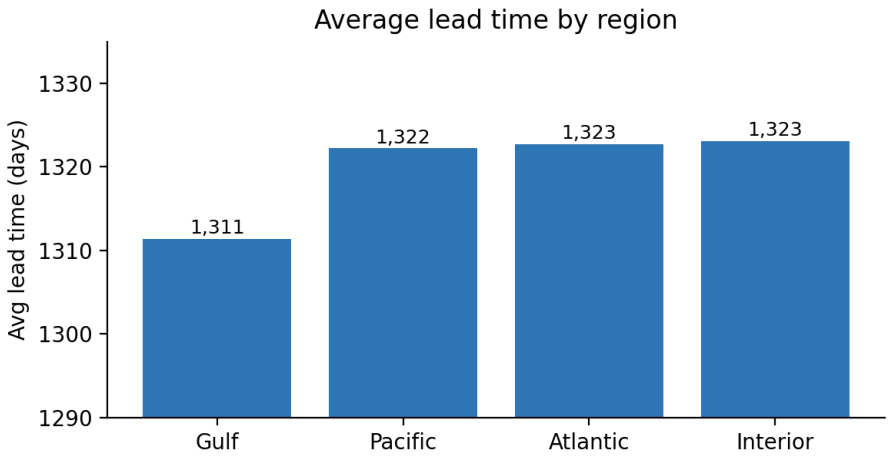


Figure 4. Average lead time by customer region (days).

Table 6. Performance by customer region

Region	Order lines	Avg lead (days)	Std dev	Delay %
Gulf	1,620	1,311.4	264.7	20.5
Pacific	3,253	1,322.2	266.5	21.2
Atlantic	2,986	1,322.7	256.5	20.9
Interior	2,335	1,323.1	262.7	21.5

6.2 States

Among the 40 states and provinces with at least 30 order lines, the fastest by raw average lead time are Nevada (1,226.8 days, 39 orders), Virginia (1,229.1 days, 224 orders), Ontario (1,247.5 days, 58 orders), Utah (1,253.0 days, 53 orders) and South Carolina (1,255.4 days, 42 orders). The slowest are Iowa (1,443.9 days, 30 orders), New Mexico (1,441.8 days, 37 orders), Tennessee (1,391.5 days, 183 orders), Indiana (1,381.5 days, 149 orders) and Washington (1,360.7 days, 506 orders). The gap between the fastest and slowest state is about 217 days, far larger than the regional gap.

The two highest-volume states, California (2,001 orders, 1,318 days) and Texas (985 orders, 1,310 days), both perform close to the overall average, so the largest customer markets are not the problem.

6.3 Congestion-prone states

We defined a congestion-prone state as one with volume at or above the median of the 40 states (117 order lines), an average lead time above the overall average (1,320.8 days) and a delay frequency above the overall rate (21.1%). On raw numbers, eight states qualify (Table 7). The last column shows the year-adjusted lead time introduced in Section 8, which is the more reliable indicator because it removes the order-year effect.

Table 7. Congestion-prone states on raw numbers, with year-adjusted view

State	Order lines	Avg lead (days)	Delay %	Year-adjusted (days)
New York	1,128	1,324.5	21.5	+1.4
Pennsylvania	587	1,324.2	22.3	+10.7
Washington	506	1,360.7	23.9	+10.5
Illinois	492	1,323.2	21.7	+0.1
Georgia	184	1,338.6	25.0	+4.3
Tennessee	183	1,391.5	31.7	+45.3
Indiana	149	1,381.5	30.9	+39.7
New Jersey	130	1,338.3	27.7	-27.8

Only Tennessee and Indiana remain clearly slower after year adjustment. Washington and Pennsylvania are only mildly slower, New York, Illinois and Georgia are essentially at the average, and New Jersey is actually faster than average once the order-year mix is removed. We therefore treat Tennessee and

Indiana as the genuine high-volume bottleneck states, and the other raw-flagged states as a watch list at most.

7. Ship Mode Performance

Table 8 and Figure 5 compare the four ship modes. Standard Class has the lowest average lead time (1,314.3 days), followed by Second Class (1,323.8), Same Day (1,333.4) and First Class (1,338.3). Standard Class is therefore about 20 days faster than the premium modes, which is the opposite of what the service names imply. Medians are nearly identical (1,269 to 1,274 days), and the spread is similar across modes (standard deviation 254 to 266 days).

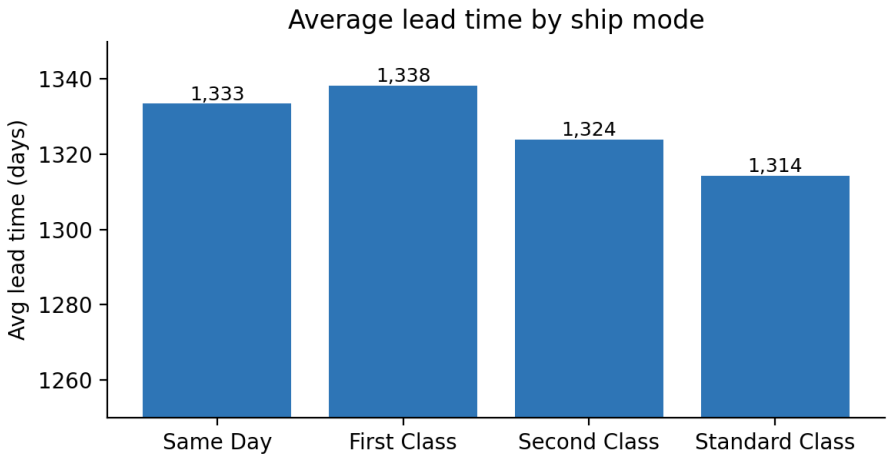


Figure 5. Average lead time by ship mode (days).

Table 8. Ship mode comparison

Ship mode	Orders	Avg lead	Median	Std dev	Delay %	Cost/unit	Margin %
Standard Class	6,120	1,314.3	1,274	262.4	30.9	1.24	66.0
Second Class	1,979	1,323.8	1,273	261.8	12.9	1.25	65.7
First Class	1,548	1,338.3	1,272	265.6	0.0	1.26	65.7
Same Day	547	1,333.4	1,269	253.8	0.0	1.23	66.1

Grouped as Standard versus Expedited, Standard has 6,120 orders, an average lead time of 1,314 days, a 31% delay frequency and a 66.0% margin, while Expedited has 4,074 orders, 1,331 days, 6% delay frequency and a 65.8% margin. Expedited orders are also not faster than Standard at any of the five factories.

7.1 Reading the delay and cost results with care

- The delay percentages are mostly an artifact of the threshold. The 75th percentile cut-off (1,638 days) sits at the top of the 2025 cluster, and only Standard Class and Second Class orders reach above it. First Class and Same Day show 0% delay simply because their longest lead times (1,638 and 1,636 days) do not exceed the threshold. Delay frequency should not be read as proof that premium modes are more reliable.

- Within each order year the four modes are almost the same: about 1,088 to 1,102 days for 2024 orders and 1,468 to 1,489 days for 2025 orders, a difference of at most about 21 days. The year-to-year gap (about 385 days) is far larger than any mode gap.
- Cost per unit (1.23 to 1.26) and profit margin (65.7% to 66.1%) hardly differ by mode. Descriptively, the data shows no measurable time benefit from the premium modes and no measurable cost difference either. This is a descriptive observation only and does not establish cause and effect.

8. Robustness Check: Year-Adjusted Lead Time

Because lead time depends on order year, a state with more 2025 orders looks slower. Across the 40 states with at least 30 orders, the correlation between a state's raw average lead time and its share of 2025 orders is 0.73 (Figure 6). To remove this effect we computed an adjusted lead time for each order: its lead time minus the average lead time of all orders placed in the same year. A state's adjusted score is the mean of these values, so a positive number means the state is slower than the same-year average and a negative number means faster.

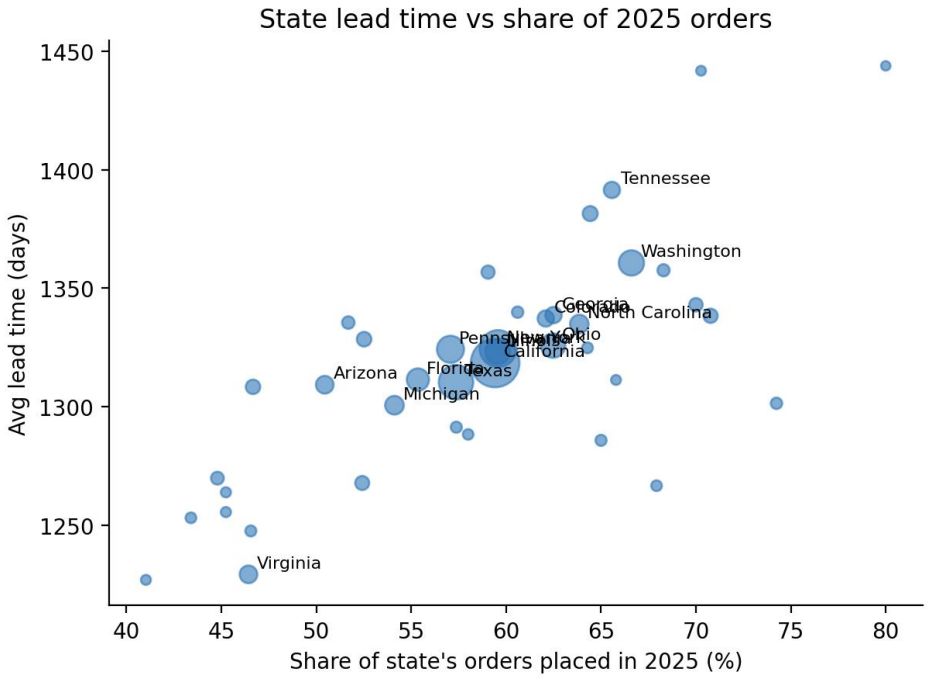


Figure 6. State average lead time rises with the share of orders placed in 2025.

After adjustment, the state-level range falls from about 217 days to about 166 days, and the correlation with the 2025 share is approximately zero by construction. Figure 7 shows the states that are most clearly faster and slower. The regional picture does not change: the adjusted values for the four regions are within about 4 days of zero (Atlantic +1.8, Gulf -3.6, Interior -0.3, Pacific +0.4).

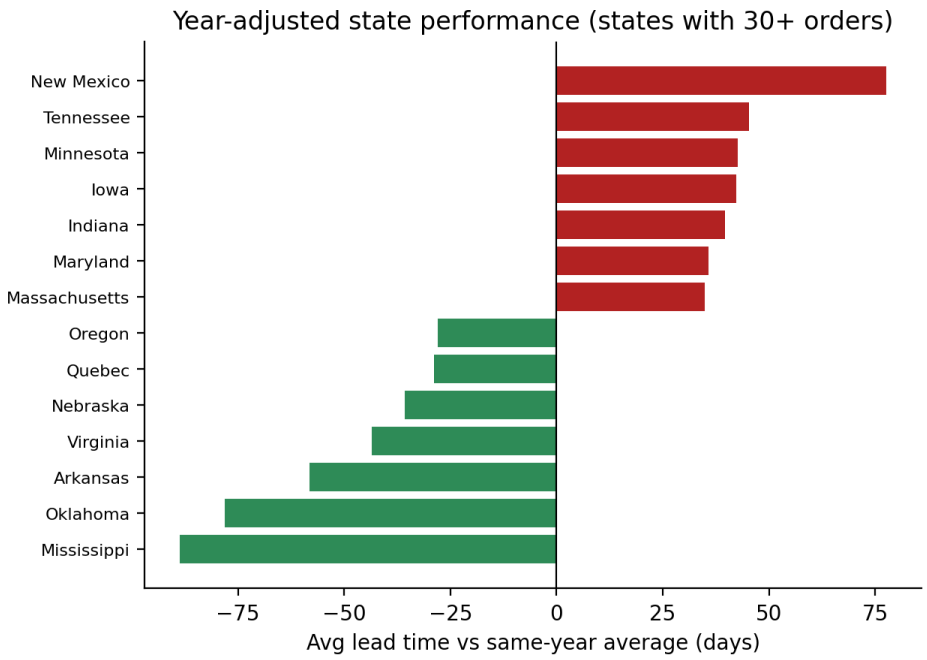


Figure 7. Year-adjusted state performance for the seven fastest and seven slowest states (30+ orders).

Key results of the adjusted view:

- **Consistently slower:** Tennessee (+45.3 days, 183 orders), Indiana (+39.7, 149 orders) and Maryland (+35.7, 105 orders) are slower under both raw and adjusted measures. Minnesota (+42.7, 89 orders), Massachusetts (+34.8, 135 orders) and Kentucky (+32.5, 139 orders) are close to average on raw numbers but clearly slower after adjustment.
- **Low-volume extremes:** New Mexico (+77.6 days, 37 orders, 56.8% delay) and Iowa (+42.2 days, 30 orders) are slow under both views, but their small order counts limit confidence.
- **Consistently faster:** Virginia (-43.4 days, 224 orders) and Oregon (-27.9, 124 orders) are faster under both views at meaningful volume. The largest adjusted advantages are in smaller states: Mississippi (-88.6, 53 orders), Oklahoma (-78.1, 66 orders) and Arkansas (-58.2, 60 orders). Nevada (-25.0, 39 orders) is also faster.
- **Largely a time-mix effect:** Washington looks slow on raw numbers (1,360.7 days) but is only +10.5 days after adjustment, because about two thirds of its orders were placed in 2025. New Jersey's raw slowness (1,338.3 days) reverses to -27.8 days.

9. Recommendations

Because the dataset's dates are not realistic, the recommendations below are about relative priorities. Each should be confirmed with real shipping and delivery dates before major spending or contract decisions.

1. **Verify the shipping date data first.** Confirm with the data owner how Ship Date is recorded and why ship dates fall years after order dates. Capture actual ship dates and, ideally, delivery dates going forward. Every other recommendation depends on this.
2. **Investigate Tennessee and Indiana first.** They are high-volume, have delay frequency near 31%, and are slower than average under both raw and year-adjusted measures. The slow routes are served by both large factories (Lot's O' Nuts and Wicked Choccy's), so the cause is more likely on the destination or carrier side than at a single factory. Review carrier handoffs and local delivery arrangements for these states.
3. **Keep a watch list.** Maryland, Massachusetts, Kentucky and Minnesota are slow after adjustment, and New Mexico and Iowa are slow with low volume. Track them monthly. Treat Washington, Pennsylvania and Georgia as lower priority because their raw slowness is mostly due to order-year mix.
4. **Learn from the fast routes.** Virginia is fast from both large factories with real volume (about 106 and 109 orders per route), and Oregon, Nevada, Mississippi, Oklahoma and Arkansas also perform well. Compare their carrier, handling and lane practices with Tennessee and Indiana.
5. **Re-examine ship mode pricing and promises.** In this data the premium modes show no time advantage, while costs and margins are similar. If the real dates confirm this, customers paying for premium modes may not be receiving faster service, and mode selection and carrier performance should be reviewed. Until then, avoid conclusions about which mode is faster.
6. **Build volume where routes are thin.** Three factories (Secret Factory, The Other Factory, Sugar Shack) ship only 350 order lines in total. Their route results are not reliable, so continue to apply a minimum order rule when ranking routes.
7. **Adopt the dashboard as a routine monitor.** Review the route leaderboard, state map and ship mode view each month, and add an order-date control (or adjustment) so that comparisons are not distorted by time effects.

10. Limitations and Future Work

- Lead times are not realistic and depend on order year, so only relative conclusions are supported.
- The delay threshold is a percentile of the data, so some share of shipments is flagged by construction, and the cut-off mainly flags 2025 orders and the slower modes.
- Factory assignment comes from the product-factory table in the project specification. Actual fulfilment locations are not in the data.
- Many state-level routes have few orders. A minimum of 10 orders was used for route ranking and 30 for state analysis, and results for small routes carry high uncertainty.
- Canadian provinces are present in the data but are not shown on the US map.
- The cost-time analysis is descriptive only and does not establish cause and effect.

- Factory coordinates were provided but distance between factory and customer was not modelled. Future work could add distance-based efficiency (days per mile), statistical tests, and a carrier dimension once real data is available.

11. Conclusion

This project built a repeatable method to measure shipping route efficiency for Nassau Candy Distributor: lead time, route aggregation, delay frequency and a normalized efficiency score, delivered through a Streamlit dashboard. The main finding is that performance differences sit at the state level rather than the region or factory level. Tennessee and Indiana are the clearest high-volume bottlenecks, while Virginia and Oregon are consistently strong at meaningful volume. Expedited shipping showed no speed advantage in this dataset.

The most important caveat is the data itself. The lead times are not realistic and depend strongly on order year, so these results show which routes and states look better or worse relative to others, not how long deliveries truly take. Fixing the shipping date data is the first step toward turning this analysis into reliable operational decisions.

Appendix A. Dashboard and Reproducibility

The Streamlit dashboard is available at https://factorytocustomershippingrouteefficiency.streamlit.app/. It has four modules: Route Efficiency Overview, Geographic Shipping Map, Ship Mode Comparison and Route Drill-Down. Filters cover order date range, region, state, ship mode, delay threshold and minimum orders per route. The dashboard displays the data limitation as a warning at the top of every page.

The analysis was done in Python with pandas, NumPy, Plotly and Streamlit. The cleaned dataset and route tables were exported to CSV files that the dashboard reads. The year-adjusted analysis in Section 8 was produced as a supplementary check to the specified methodology.

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